

# Rakshak

## Accident Emergency & Aftermath - Build Guide

*Build It track | Best UI | 4-day MVP*

## 1) Vision & Scope

Goal: one app that helps ordinary people in India in two critical moments.

- Immediately after seeing a road accident, as a bystander.
- In the days and weeks after their own family meets with an accident.

Hackathon positioning

- Track: Build It (open-source AWS stack, running locally).
- Also competing for: Best UI.
- Story: "One app, two modes: first minutes matter, and the long road after."

MVP scope (4 days)

- Two main tabs: Emergency (bystander) and Aftermath (family).
- AI assistant (Strands Agents) for first-aid guidance and process FAQs.
- Local serverless backend: SAM + LocalStack, DynamoDB, OpenSearch.
- Cedar policies: anyone reads, only admins write.
- Panic-friendly, mobile-first UI.

## 2) User Journeys

### Journey A - 'I just saw an accident' (Emergency tab)

1. App opens on the Emergency tab by default.
2. Three full-width action buttons: Call Ambulance (108/112), Call Police (112), Share Location & Info (pre-filled WhatsApp/SMS with a Maps link).
3. Quick triage: "Is the person breathing?" -> Yes / No / Not sure.
  - Not breathing or not sure: check responsiveness, call 112, start chest compressions at 100-120/min, continue until help arrives.
  - Breathing: do not move them unless there is fire risk, control bleeding with firm pressure, keep them warm and still, no food or water.
4. More help: nearest trauma centres, Good Samaritan rights in plain language.
5. Ask-a-question box backed by EmergencyGuideAgent.

Microcopy that matters

- "You don't need to be a doctor. Just follow these steps."
- "You are protected as a Good Samaritan. You don't have to reveal your identity."

### Journey B - 'My family met with an accident' (Aftermath tab)

Four stages, each a card with a checklist plus short explanations.

At Hospital

- Ensure the victim is registered as an MLC (Medico-Legal Case).
- Collect admission summary, prescriptions, test reports, every bill.
- Emergency care cannot be denied for lack of payment or police formalities.

At Police Station

- File the FIR at the station nearest the accident spot.
- Check that time, place, vehicles and injuries are recorded correctly.

- Always ask for a copy of the FIR; insist on hit-and-run registration if relevant.

#### Insurance Claim

- Inform the insurer as early as possible.
- Assemble FIR copy, MLC papers, bills, discharge summary, photographs.
- Hit-and-run cases: the Solatium Fund route via the district administration.

#### Legal Aid & Compensation

- Motor Accidents Claims Tribunal (MACT) handles compensation claims.
- Free legal aid through State Legal Services Authorities (SLSA).
- Claims can cover medical costs, lost income, disability and death.

### 3) Tech Architecture

- Frontend: React / Next.js, two tabs, mobile-first.
- AI: Strands Agents SDK - EmergencyGuideAgent and AftermathGuideAgent.
- Backend: SAM CLI + LocalStack. API Gateway -> Lambda -> DynamoDB + OpenSearch.
- Tables: EmergencyProtocols, AftermathSteps, Resources, FAQs.
- Search: OpenSearch indexes for resources and aftermath topics.
- Authorization: Cedar policies evaluated inside the Lambda handlers.

### 4) Data Models

EmergencyProtocols scenario\_id, title, steps[], do\_not[], tags[]

AftermathSteps stage\_id, title, checklist[], details[], tags[]

Resources resource\_id, name, type, state, city, address, contact, description, tags[]

FAQs faq\_id, question, answer, topic, tags[]

```
{
  "scenario_id": "not_breathing",
  "title": "Person is not breathing",
  "severity": "critical",
  "steps": [
    "Tap the shoulder and shout - check if they respond.",
    "Call 112 now. Put the phone on speaker.",
    "If trained, push hard and fast in the centre of the chest, 100-120/min.",
    "Keep going until they breathe or the ambulance arrives."
  ],
  "do_not": ["Do not give water.", "Do not shake the head or neck."],
  "tags": ["breathing", "cpr", "critical"]
}
```

### 5) Strands Agents Design

EmergencyGuideAgent tools

- get\_emergency\_protocol(scenario)
- get\_state\_emergency\_numbers(state)
- get\_good\_samaritan\_info()

#### AftermathGuideAgent tools

- get\_aftermath\_steps(stage)
- search\_resources(query, state, type)
- answer\_aftermath\_faq(question)

Keep the emergency agent deterministic: short numbered steps, no speculation, and always end with "Call 112 if you have not already."

## 6) Backend APIs

GET /emergency/protocols?scenario=not\_breathing

GET /emergency/good-samaritan

GET /aftermath/steps?stage=hospital

GET /aftermath/faqs?topic=fir&q=delay

GET /resources?type=hospital&state=Karnataka&city=Bengaluru

POST /ai/emergency

POST /ai/aftermath

Full request and response shapes are in docs/api-spec.md; seed payloads for every table are in data/\*.json.

## 7) Cedar Policies

```
// Anyone may read the public safety content
permit(
  principal,
  action in [Action::"read", Action::"search"],
  resource in Namespace::"PublicContent"
);

// Only admins may change it
permit(
  principal in Role::"admin",
  action in [Action::"write", Action::"delete"],
  resource in Namespace::"PublicContent"
);
```

## 8) Frontend & Best UI

#### Global principles

- Mobile-first, 56-72px tall primary buttons, thumb-reachable.
- Panic-friendly: one decision per screen, minimal text, high contrast.
- Calm tone. Short sentences. No jargon without a one-line explanation.
- India-specific: 108/112, MLC, FIR, Good Samaritan, SLISA, MACT.

#### Emergency tab

- Three stacked action buttons above the fold, then the triage question, then step cards revealed by the answer.
- Red is reserved for the ambulance and police actions only - everything else is

neutral, so the critical action never competes for attention.

Aftermath tab

- Search bar, four stage cards, checklists with persisted checkbox state, and a resource list filtered by type, state and city.

## 9) 4-Day Build Plan

Day 1 Content for 4 scenarios, 4 stages, 15 FAQs. SAM + LocalStack up, tables created and seeded, first read Lambdas working.

Day 2 Strands agents, /ai endpoints, OpenSearch resource search, Cedar checks.

Day 3 Frontend: two tabs, triage flow, checklists, FAQ search, AI chat wired up.

Day 4 UI polish, sample resources, Good Samaritan screen, 3-minute demo video, README and submission write-up.

## 10) What to Emphasize for Judges

- Impact: India's accident crisis, the first minutes, and the months after.
- Stack: Strands Agents, SAM + LocalStack, OpenSearch, Cedar - all open source.
- Learning: first agentic app, first serverless app, first Cedar policies.
- Execution: both journeys work end to end, not just the demo path.
- Demo: show real use, not architecture diagrams. Lead with the crash scene.